In vitro investigation of the potential immunomodulatory and anti-cancer activities of black pepper (Piper nigrum) and cardamom (Elettaria cardamom...

Although the immunomodulatory effects of many herbs have been extensively studied, research related to possible immunomodulatory effects of various spices is relatively scarce. Here, the potential immunomodulatory effects of black pepper and cardamom are investigated. Our data show that black pepper and cardamom aqueous extracts significantly enhance splenocyte proliferation in a dose-dependent, synergistic fashion. Enzyme-linked immunosorbent assay experiments reveal that black pepper and cardamom significantly enhance and suppress, respectively, T helper (Th)1 cytokine release by splenocytes. Conversely, Th2 cytokine release by splenocytes is significantly suppressed and enhanced by black pepper and cardamom, respectively. Experimental evidence suggests that black pepper and cardamom extracts exert pro-inflammatory and anti-inflammatory roles, respectively. Consistently, nitric oxide production by macrophages is significantly augmented and reduced by black pepper and cardamom, respectively. Remarkably, it is evident that black pepper and cardamom extracts significantly enhance the cytotoxic activity of natural killer cells, indicating their potential anti-cancer effects. Our findings strongly suggest that black pepper and cardamom exert immunomodulatory roles and antitumor activities, and hence they manifest themselves as natural agents that can promote the maintenance of a healthy immune system. We anticipate that black pepper and cardamom constituents can be used as potential therapeutic tools to regulate inflammatory responses and prevent/attenuate carcinogenesis.